清華大學机械工程系
Department of Mechanical Engineering, Tsinghua University

Exploratory Experiment 1:
Study on the properties of various knives

Various knives are used in our daily life. Sharpness, appearance, and durability are the main requirements for the household knives. For this topic, please complete the following exploratory research:

1. Purchase the knife you selected, and prepare the test samples; you can choose more than one knives for comparison;
2. Observe the microstructure, analyze the chemical composition, and measure the hardness of the material used to make a knife;
3. Analyze the relationship between the microstructure and the properties.

Submission:

1. A research report (font size 12; about 10 pages of A4 paper)
2. Oral presentation with PowerPoint, time 12 minutes

You are free to choose any other types of tools/materials for this work.
Keep the receipts for reimbursement.

Exploratory Experiment 2: Looking for the hardest metal in life



There are many metal products in our daily life. Their hardnesses are different for different applications. Please try to find the "hardest" metal in your life, and conduct a comparative analysis on the metal products that you bought to find the causes of the high hardness:

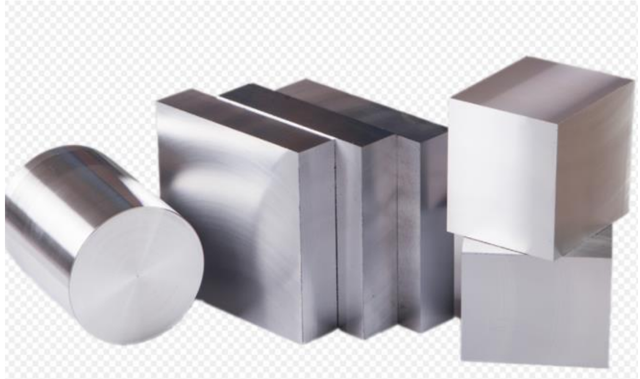
1. Purchase the metal products you choose, and prepare the test samples;
2. Measure the hardness. A variety of different hardness comparison methods can be used (hardness tester, metallographic observation after mutual marking, etc.)
3. Analyze the microstructure characteristics of the metal with the highest hardness, and find the possible causes of its high hardness.

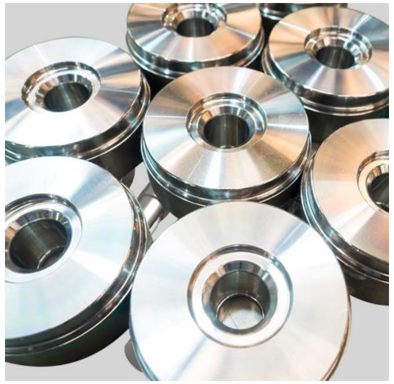
Submission:

1. A research report (font size 12; about 10 pages of A4 paper)
2. Oral presentation with PowerPoint, time 12 minutes

At least one laboratory instrument should be used (microscope, hardness tester, heat treatment furnace, etc.); Keep the receipts for reimbursement.

Exploratory Experiment 3:
Identification of common metal materials





Recently, several different metallic samples were mixed in a lab maintenance work. Although the types of these samples are known, the samples cannot be identified just by their appearance. Please distinguish these samples by using the knowledge learned in the course:

1. The mixed samples will be provided by the laboratory;
2. The samples need be distinguished by various examination methods, such as microscope and hardness tester.;

Submission:

1. A research report (font size 12; line spacing 1.2; about 10 pages of A4 paper)
2. Oral presentation with PowerPoint, time 12 minutes

At least one laboratory instrument should be used (microscope, hardness tester, heat treatment furnace, etc.).